Defective Analysis

Framing, Terminology, and Problem Description

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Postulated analysis needing confirmation and review

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PDF 1 of 4. This draft defines the top-level framing of Defective Analysis before introducing Defective Calculus, graph examples, or COTG realizations.

Claim boundary. This document defines a proposed analytic vocabulary. It uses ordinary mathematical language: set, class, map, equivalence relation, projection, fiber, kernel, quotient, section, residual, admissibility, and clock signature. The symbols θ and α are local notation for two operation modes; they are not asserted to be new mathematical structures. The local/global classifier below is presented as a postulated diagnostic rule, not as a completed theorem. No claim is made about external applications, physical systems, or open mathematical conjectures.

Abstract

Defective Analysis (DA) is a proposed framework for describing problems by assigning an operation clock, identifying the data exposed by that clock, and stating what remains hidden in the corresponding fiber. The first diagnostic task in DA is clock assignment. A problem is locally determined when it is governed by exactly one clock: either a θ-clock or an α-clock. A problem is globally coupled when both clocks are required. A problem is malformed when the assigned clock signature is wrong. This document fixes the basic terminology and gives a protocol for describing a problem without smuggling in side information.

1 Purpose

Defective Analysis is the top-level framework. It contains, but is not identical with, Defective Calculus, Whole Defect Systems, Scalar Calculus, and Charted Oscillatory Transport Geometry. Its purpose is to describe a problem before attempting to solve it.

The basic DA question is not immediately "what is the answer?" It is:

which clock structure makes the problem well-formed?

Once the clock structure is assigned, DA asks what a legal projection exposes, what the corresponding reconstruction must retain, and whether the remaining ambiguity is admissible or the result of a wrong assignment.

2 Primitive terms

A DA description of a problem begins with the following data.

Symbol	Meaning
P	The problem being described.
$\mathcal{X}$	A carrier class of admissible objects or states.
$\sim$	The declared equivalence relation on candidate solutions.
θ	The derivation/projection/probe clock.
α	The integration/realization/reconstruction clock.
D_θ	The defective observable generated by the θ-clock.
I_α	The reconstruction or realization operation generated by the α-clock.
$\mathcal{F}_\theta$	The fiber of states sharing the same defective observable.
K_{ret}	The admissible retained-kernel class needed for reconstruction.
Π	The visible quotient or visible-data projection.

σ	The polarity state; $\sigma = \perp$ means unresolved polarity.
Adm	The admissibility rule preventing leakage and invalid reconstruction.

A clock is not a literal time variable. A clock is an allowed mode of operation. A θ-clock exposes, deletes, differentiates, probes, projects, or coarse-grains. An α-clock glues, integrates, completes, realizes, evolves, or reconstructs.

3 Clock assignment

Definition 1 (Clock signature)**.** *The clock signature of a problem P is a subset*

$$\mathrm{Clk}(P) \subseteq \{\theta, \alpha\}.$$

It records which operation clocks are necessary for the problem to be well-formed.

Postulate 1 (Local/global diagnostic)**.** *The first DA diagnostic is the following clock-classification rule:*

$$\textit{local} = \theta \oplus \alpha,$$

$$\textit{global} = \theta \wedge \alpha,$$

$$\textit{malformed} = \textit{wrong clock assignment}.$$

Here "local" means single-clock deterministic. It does not mean spatially small. A local problem is governed by exactly one clock. A global problem requires both clocks and therefore requires their interaction, order, compatibility, or closure. A malformed problem is one whose assigned clock signature does not match the operations that actually make it well-formed.

Thus DA distinguishes four cases:

$$\mathrm{Class}_{\mathrm{DA}}(P) = \begin{cases} \theta\text{-local}, & \mathrm{Clk}(P) = \{\theta\}, \\ \alpha\text{-local}, & \mathrm{Clk}(P) = \{\alpha\}, \\ \theta\alpha\text{-global}, & \mathrm{Clk}(P) = \{\theta, \alpha\}, \\ \text{malformed}, & \mathrm{Clk}_{\mathrm{assigned}}(P) \neq \mathrm{Clk}_{\mathrm{true}}(P). \end{cases}$$

4 Problem-description protocol

A DA problem description should state the following items before any attempted solution.

1. **Carrier.** Declare the object class $\mathcal{X}$.
2. **Equivalence.** Declare $\sim$, the relation under which two solutions count as the same.
3. **Given data.** Declare what data are actually supplied.
4. **Legal operations.** Declare the legal θ- and α-operations.
5. **Clock signature.** State whether the problem is θ-local, α-local, $\theta\alpha$-global, or currently malformed.
6. **Projection.** If present, define $D_\theta : \mathcal{X} \to \mathcal{Y}$.
7. **Fiber.** Define the unresolved fiber $D_\theta^{-1}(D_\theta X)$.
8. **Retained kernel.** State what class of hidden data is admissible to retain.
9. **Polarity.** State whether the polarity is determined or unresolved.

10. **No-leakage rule.** State what information cannot be introduced without invalidating the problem.

This protocol is intended to keep the framework reviewable. A problem may be expressive and still be invalid if its solution uses data that were not supplied or legally generated.

5 Local, global, and malformed problems

A θ-local problem is deterministic under projection, differentiation, deletion, or probing alone. An α-local problem is deterministic under reconstruction, completion, gluing, or realization alone. A $\theta\alpha$-global problem is one in which the interaction of projection and reconstruction is essential.

A typical global obstruction has the form

$$D_\theta I_\alpha \neq I_\alpha D_\theta,$$

when both composites are meaningful. The difference, mismatch, or failure of these operations to close is the defect residual.

The central failure mode is not mystery. It is wrong assignment:

$$\mathrm{Clk}_{\mathrm{assigned}}(P) \neq \mathrm{Clk}_{\mathrm{true}}(P).$$

A wrong clock assignment can create artificial defects. DA therefore requires clock diagnosis before interpreting an obstruction as intrinsic.

6 Static and operational regimes

DA separates two regimes that must not be conflated.

Definition 2 (Static regime). *In a static regime, the available information is only the declared observable $D_\theta(X)$. A scalar response, operator action, or hidden support may be used only if it is already encoded in the declared observable. Otherwise it is leakage.*

Definition 3 (Operational regime). *In an operational regime, the object is hidden but can be queried through declared measurement ports and legal clock-generated operations. A probe is legal only if it is produced from an allowed port through an admissible α, θ sector.*

If $\mathcal{P}$ is the declared measurement-port subspace and $\mathcal{O}_{\alpha,\theta} = \langle \alpha, \theta \rangle$ is the legal operation algebra, then the legal probe space is

$$\mathcal{Z}_{\mathrm{legal}} = \mathrm{Sector}_{\alpha,\theta}(\mathcal{P}).$$

In an operational regime, one may measure scalar responses of legal probes. In a static regime, the same response is inadmissible unless it is part of the data.

7 No-leakage

Definition 4 (No-leakage). *A DA argument satisfies no-leakage when every quantity used in the solution is either contained in the stated data or generated by a declared legal operation. If a quantity identifies the hidden representative, hidden support, or missing reconstruction choice without being legal data, the argument leaks.*

No-leakage is the discipline that distinguishes a legitimate retained kernel from an arbitrary rescue channel.

8 Relation to subordinate documents

Defective Calculus is the subcalculus of D_θ, I_α, fibers, retained kernels, residuals, and admissibility.

A Whole Defect System is an upstream object in which a carrier is equipped with a defect extractor, defect sectors, a gluing carrier, and a quotient/gluing projection.

COTG is a charted oscillatory realization layer derived through a Whole Defect System path. It supplies local charts, transport, tensor or completed state objects, projections, residuals, and intrinsic admissibility.

Scalar Calculus is a finite quotient and parity-kernel model used to test that the grammar is nonempty, constrained, and falsifiable.

9 Failure surfaces

Code	Failure surface
F1	Wrong clock assignment: the problem is treated as θ-local, α-local, or global incorrectly.
F2	Leakage: the solution uses data not supplied or legally generated.
F3	Undefined equivalence: the relation $\sim$ is not stated.
F4	Arbitrary retained kernel: hidden data are added without an admissibility rule.
F5	Projection/reconstruction mismatch: D_θ and I_α do not act in compatible categories or carriers.
F6	False polarity: $\sigma = \perp$ is collapsed without a legal resolving operation.
F7	Static/operational conflation: a probe response is used in a static problem without an oracle or declared port.

10 Known unknowns and proof obligations

1. Define categories of DA problem descriptions and morphisms between them.
2. Specify when the true clock signature can be derived rather than assigned.
3. State conditions under which polarity is derived from a fiber and when it must be primitive.
4. Formalize legal probe generation in operational regimes without smuggling hidden representatives.
5. Relate DA failure surfaces to standard mathematical counterexamples and collapse tests.

11 Final compression

Defective Analysis is the proposed framework for assigning clock signatures to problems and then studying the observable, fiber, retained kernel, polarity, and admissibility induced by that assignment.

$$\boxed{\mathrm{DA}(P) = (\mathrm{Clk}(P), D_\theta, I_\alpha, \mathcal{F}_\theta, K_{\mathrm{ret}}, \Pi, \sigma, \mathsf{Adm}).}$$

$$\boxed{\text{local} = \theta \oplus \alpha, \quad \text{global} = \theta \wedge \alpha, \quad \text{failure} = \text{wrong clock assignment}.}$$